

Program 1: Addition of Two 8-bit Number

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	00
BC	00 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00 Update Port Value

Memory

0 - + 00 Update Memory

Load me at

```
1 LDA 8050H
2 MOV B,A
3 LDA 8051H
4 ADD B
5 STA 8052H
6 HLT
```

Start 8050

Address (Hex)	Address	Data
1F72	8050	5
1F73	8051	3
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	08
BC	05 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00 Update Port Value

Memory

0 - + 00 Update Memory

Load me at

```
1 LDA 8050H
2 MOV B,A
3 LDA 8051H
4 ADD B
5 STA 8052H
6 HLT
```

Start 8050h

Address (Hex)	Address	Data
8050	32848	5
8051	32849	3
8052	32850	8
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

2. Subtraction of Two 8-bit Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center pane is as follows:

```
1 LDA 8050H
2 MOV B,A
3 LDA 8051H
4 MOV C,A
5 MOV A,B
6 SUB C
7 STA 8052H
8 HLT
```

The registers pane on the left shows the following values:

Register	Value
A	02
BC	05 03
DE	00 00
HL	00 00
PSW	00 00
PC	42 0E
SP	FF FF
Int-Reg	00

The flag pane shows the following values:

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

The memory pane on the right shows the following data:

Address (Hex)	Address	Data
8050	32848	5
8051	32849	3
8052	32850	2
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message pane at the bottom shows: "Program assembled successfully".

3. Multiplication of Two 8-bit Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code in the center pane is as follows:

```
1 LDA 8050H
2 MOV B,A
3 LDA 8051H
4 MOV C,A
5 MVI A,00H
6 LOOP: ADD B
7 DCR C
8 JNZ LOOP
9 STA 8052H
10 HLT
```

The registers pane on the left shows the following values:

Register	Value
A	06
BC	02 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 13
SP	FF FF
Int-Reg	00

The flag pane shows the following values:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

The memory pane on the right shows the following data:

Address (Hex)	Address	Data
8050	32848	2
8051	32849	3
8052	32850	6
8053	32851	0
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

The assembler message pane at the bottom shows: "Program assembled successfully".

Program 4: Division of Two 8-bit Numbers

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag
A	04	S 1
BC	04 05	Z 0
DE	02 00	AC 0
HL	00 00	P 1
PSW	00 00	C 1
PC	42 1E	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

0 - + 00

Update Memory

Load me at

```
1 LDA 8050H
2 MOV B,A
3 LDA 8051H
4 MOV C,A
5
6 MVI D,00H
7
8 LOOP: MOV A,B
9 CMP C
10 JC EXIT
11 SUB C
12 MOV B,A
13 INR D
14 JND LOOP
15
16 EXIT: MOV A,D
17 STA 8052H
18 MOV A,B
19 STA 8053H
20 HLT
```

Start 8050h

Address (Hex)	Address	Data
8050	32848	14
8051	32849	5
8052	32850	2
8053	32851	4
8054	32852	0
8055	32853	0
8056	32854	0
8057	32855	0
8058	32856	0
8059	32857	0
805A	32858	0
805B	32859	0
805C	32860	0
805D	32861	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

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